

Homework Set HW6 due 3/1/04 at 12:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt) Problem 3 Section 3.5 page 143.

If $y = (u^2 + 1)$ and $u = 3x - 2$, use the chainrule to find dy/dx and write your answer in terms of x only.

2.(1 pt) Problem 4 Section 3.5 page 143.

If $y = (2u^2 - u + 5)$ and $u = 1 - x^2$, use the chainrule to find dy/dx and write your answer in terms of x only.

3.(1 pt) This is problem 13, Section 3.5 page 143.

If $f(x) = (5x - 2)^5$, find $f'(x)$.

4.(1 pt) This is problem 14, Section 3.5 page 143.

If $f(x) = (x^4 - 7x)^{15}$, find $f'(x)$.

5.(1 pt) This is problem 16, Section 3.5 page 143.

If $f(x) = (3 - x^2 - x^4)^{11}$, find $f'(x)$.

6.(1 pt) This is problem 18, Section 3.5 page 143.

If $f(x) = \cos(5 - 3x)$, find $f'(x)$.

7.(1 pt) This is problem 20, Section 3.5 page 143.

If $f(x) = e^{x^3 - \pi}$, find $f'(x)$.

8.(1 pt) This is problem 22, Section 3.5 page 143.

If $g(x) = e^{\sin(x)}$, find $g'(x)$.

9.(1 pt) This is problem 24, Section 3.5 page 143.

If $g(t) = t^2 e^{-t} + (\ln(t))^2$, find $g'(t)$.

10.(1 pt) This is problem 26, Section 3.5 page 143.

If $h(x) = \frac{\tan(3x)}{x^2}$, find $h'(x)$.

11.(1 pt) This is problem 30, Section 3.5 page 143.

If $g(x) = \ln(3x^4 + 5x)$, find $g'(x)$.

12.(1 pt) This is problem 32, Section 3.5 page 143.

If $f(x) = \csc^2(\sqrt{x})$, find $f'(x)$.

13.(1 pt) This is problem 41, Section 3.5 page 143.

Find the x -coordinate of each point where the graph of the function

$$f(x) = x\sqrt{1 - 3x}$$

has a horizontal tangent line.

The sum of these x coordinates is _____

14.(1 pt) This is problem 42, Section 3.5 page 143.

Find the x -coordinate of each point where the graph of the function

$$f(x) = x^2(2x + 3)^2$$

has a horizontal tangent line.

The sum of these x coordinates is _____

15.(1 pt) This is problem 44, Section 3.5 page 143.

Find the x -coordinate of each point where the graph of the function

$$f(x) = (2x^2 - 7)^3$$

has a horizontal tangent line.

The sum of these x coordinates is _____

16.(1 pt) This is problem 46, Section 3.5 page 143.

Find the x -coordinate of each point where the graph of the function

$$f(x) = \frac{\ln(\sqrt{x})}{x^2}$$

has a horizontal tangent line.

The sum of these x coordinates is _____

17.(1 pt)

This is problem 1 Section 3.6 page 155.

Use implicit differentiation to $y' = dy/dx$ given

$$x^2 + y^2 = 25$$

$y' =$ _____

18.(1 pt)

This is problem 2 Section 3.6 page 155.

Use implicit differentiation to $y' = dy/dx$ given

$$x^2 + y = x^3 + y^3$$

$y' =$ _____

19.(1 pt)

This is problem 4 Section 3.6 page 155.

Use implicit differentiation to $y' = dy/dx$ given

$$xy(2x + 3y) = 2$$

$y' =$ _____

20.(1 pt)

This is problem 6 Section 3.6 page 155.

Use implicit differentiation to $y' = dy/dx$ given

$$x^3 + y^3 = x + y$$

$y' =$ _____

21.(1 pt)

This is problem 19 Section 3.6 page 155.

Find dy/dx given

$$y = \sin^{-1}(2x + 1)$$

$y' =$ _____

22.(1 pt)

This is problem 20 Section 3.6 page 155.

Find dy/dx given

$$y = \cos^{-1}(4x + 3)$$

$y' =$ _____

23.(1 pt)

This is problem 33 Section 3.6 page 155.

Use implicit differentiation to find the slope of the tangent line to the curve

$$x^2 + y^2 = 13$$

at the point $(-2, 3)$ and use it to find the equation of the tangent line in the form $y = mx + b$.

$$y = \underline{\hspace{2cm}}$$

24.(1 pt)

This is problem 34 Section 3.6 page 155.

Use implicit differentiation to find the slope of the tangent line to the curve

$$x^3 + y^3 = y + 21$$

at the point $(3, -2)$ and use it to find the equation of the tangent line in the form $y = mx + b$.

$$y = \underline{\hspace{2cm}}$$

25.(1 pt)

This is problem 36 Section 3.6 page 155.

Use implicit differentiation to find the slope of the tangent line to the curve

$$3^x + \log_2(xy) = 10$$

at the point $(2, 1)$ and use it to find the equation of the tangent line in the form $y = mx + b$.

$$y = \underline{\hspace{2cm}}$$

26.(1 pt) Use implicit differentiation to find the slope of the tangent line to the curve

$$xy^3 + xy = 12$$

at the point $(6, 1)$.

$$m = \underline{\hspace{2cm}}$$

27.(1 pt)

This is problem 45 Section 3.6 page 155.

Use implicit differentiation to find the second derivative y'' given

$$7x + 5y^2 = 1$$

$$y'' = \underline{\hspace{2cm}}$$

28.(1 pt)

This is problem 46 Section 3.6 page 155.

Use implicit differentiation to find the second derivative y'' given

$$x^2 + 2y^3 = 4$$

$$y'' = \underline{\hspace{2cm}}$$

29.(1 pt)

This is problem 50 Section 3.6 page 155.

Use logarithmic differentiation to find the derivative y' of

$$y = \sqrt[18]{(x^{10} + 1)^3(x^7 - 3)^8}$$

Your answer can be written as $y' = yA(x)$, find $A(x)$

$$A(x) = \underline{\hspace{2cm}}$$

30.(1 pt)

This is problem 52 Section 3.6 page 155.

Use logarithmic differentiation to find the derivative y' of

$$y = \frac{e^{2x}}{(x^2 - 3)^2 \ln(\sqrt{x})}$$

Your answer can be written as $y' = yA(x)$, find $A(x)$

$$A(x) = \underline{\hspace{2cm}}$$